

No-Compromise Uncertainty Quantification in a Stable Machine Learning Emulator of Subgrid Climate Physics

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Key Points:

- Using a confidence loss allows us to generate well-calibrated estimates of uncertainty at inference time.
- Incorporating a confidence loss does not come with sacrifices to training dynamics, offline skill, or online accuracy and stability.
- The confidence loss allows us to assess difficult-to-emulate convective regimes without the use of an expensive reference simulation.

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